A Heterogeneous Federated Learning Approach for Decentralized Pest Identification under Non-IID Distributions

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***Abstract*—Federated Learning (FL) enables decentralised model training without exchanging raw data, yet practical deployments face two intertwined obstacles: non-IID data distributions that drive client gradient updates to diverge, and gradient-based property inference attacks that allow a server-side adversary to infer the statistical composition of each client's private dataset from shared model updates. We propose a heterogeneous FL framework for fine-grained agricultural pest classification on the IP102 benchmark (α=0.5 Dirichlet partition, 102 subclasses, 8 crop-host superclasses) in which six clients deploy structurally distinct backbone architectures-EfficientNet-B0, ResNet-50, MobileNetV3-Large, ConvNeXt-Tiny, DenseNet-121, and ConvNeXt-Small-whose dimensional incompatibility renders cosine-similarity property inference attacks mathematically undefined without any noise injection or cryptographic mechanism. A server-maintained 7,418-sample clustered spatial buffer, constructed via Hierarchical Agglomerative Clustering (HAC) over backbone feature centroids, generates class-prior soft-label vectors broadcast to clients each round as a knowledge distillation (KD) signal, with the distillation weight and temperature progressively scheduled over 100 federation rounds. The proposed system achieves 63.76% test accuracy (macro-F1 = 0.611) on IP102, an 18.75 pp gain over the server-only baseline and only 1.99 pp below the best homogeneous FL configuration. Non-buffer class accuracy improves by 12.57 pp-6.7$\times$ the buffer-class gain-confirming genuine cross-class coordination, while property inference attack accuracy is reduced from 56.7-61.7% under homogeneous FL to 0% across every client and every round, demonstrating that structural privacy and competitive accuracy are simultaneously achievable.**

***Index Terms*—Federated Learning, Heterogeneous Networks, Non-IID Data, Knowledge Distillation, Gradient-Based Privacy Attacks, Architectural Incommensurability, Hierarchical Agglomerative Clustering, Agricultural Pest Identification**

I. Introduction

The rapid proliferation of edge devices and increasingly strict data privacy regulations have reshaped distributed machine learning. Federated Learning (FL) has emerged as the leading paradigm that allows multiple clients to jointly train a global model while keeping raw data local. In agricultural settings, this translates to a network of farms, research stations, and monitoring services contributing to a shared pest-classification model without centralising sensitive crop data.

The canonical FL algorithm, FedAvg, aggregates local parameters through a weighted average proportional to dataset sizes. Its fundamental assumption-that all clients deploy identical backbone architectures-is routinely violated in practice, since real edge deployments span GPU-equipped servers down to severely resource-constrained embedded devices [1]. Statistical heterogeneity compounds the problem: under realistic Dirichlet partitioning ($\alpha = 0.5$), a client representing a rice-farming region may hold predominantly rice-pest images while another holds mostly corn-pest images, causing locally trained models to diverge severely [2].

A third, less frequently co-addressed challenge is the privacy vulnerability of standard FL. When all clients share the same backbone, their gradient updates occupy the same high-dimensional parameter space, allowing a server-side adversary to compute cosine similarities between each client's update and class-specific reference gradients to identify the dominant crop type in that client's training data-a *property inference attack*. As we demonstrate empirically, this attack achieves 56.7-61.7% accuracy across three homogeneous backbones against a 12.5% random baseline for eight superclasses [3].

This paper proposes a framework that addresses all three challenges simultaneously on the IP102 benchmark [4]-102 fine-grained pest subclasses across 8 crop-host superclasses. The principal contributions are:

- **Structural privacy through heterogeneity:** six clients with structurally distinct backbones occupy incommensurable parameter spaces, reducing property inference attack accuracy from 56.7-61.7% to **0%** without differential privacy or cryptography.

- **Server-retained clustered buffer with class-prior KD:** a 7,418-sample buffer, constructed offline via HAC of backbone feature centroids and held exclusively at the server, improves non-buffer class accuracy by 12.57 pp (6.7× the buffer-class gain), confirming cross-class coordination.
- **Progressive distillation schedule:** a linear λ ramp (0.30→0.90) with temperature annealing (4.0→2.5) stabilises private-data feature learning before distillation dominates, contributing a further 5.10 pp non-buffer gain.
- **Non-buffer cross-entropy amplification:** $\gamma = 1.5$ on non-buffer subclass loss counteracts the gradient imbalance between the 32 KD-supervised and 70 CE-only subclasses.

II. Related Work

A. Federated Learning and Data Heterogeneity

Li et al. identify non-IID data and system heterogeneity as the primary bottlenecks for practical FL deployment [5]. FedProx extends FedAvg with a proximal term that penalises local updates diverging from the global state [6]. Shi et al. proposed a three-operator ADMM framework demonstrating faster convergence and robustness to client dropouts [7]. Kairouz et al. catalogue open problems in FL, including the co-design of privacy, accuracy, and efficiency-the motivation for this work [8].

B. Knowledge Distillation for Model Heterogeneity

Knowledge distillation (KD) is the dominant solution for model-heterogeneous FL, transferring knowledge through soft predictions rather than parameter averaging [9]. Zhu et al. introduced data-free generators producing synthetic samples for knowledge extraction, reaching 85.2% on CIFAR-10 without any shared dataset [10]. Shao et al. used density-ratio estimation at clients and entropy-based filtering at the server, reducing communication costs by 50-70% [11]. Li et al. showed feature distillation via orthogonal projection outperforms logit-based methods by up to 16.09% [12]. Wang et al. eliminated the need for public datasets through uncertainty-based asymmetrical reciprocity learning, outperforming prior heterogeneous FL by 5-12% [13]. The proposed system differs by using a *class-averaged* soft label from a server-retained buffer, compressing the distillation payload by 231× relative to per-image distillation.

C. Privacy in Federated Learning

Differential privacy (DP) injects calibrated noise into gradients for formal (ε, δ)-guarantees, but introduces an accuracy-privacy tradeoff that grows with non-IID severity [14]. Lomurno and Matteucci showed synthetic data generation can replace raw data sharing at compression ratios up to 600:1 [15]. Xu et al. demonstrated attribute reconstruction attacks that infer sensitive properties from shared updates even without raw data access [16]. Jin et al. combined cryptographic secure

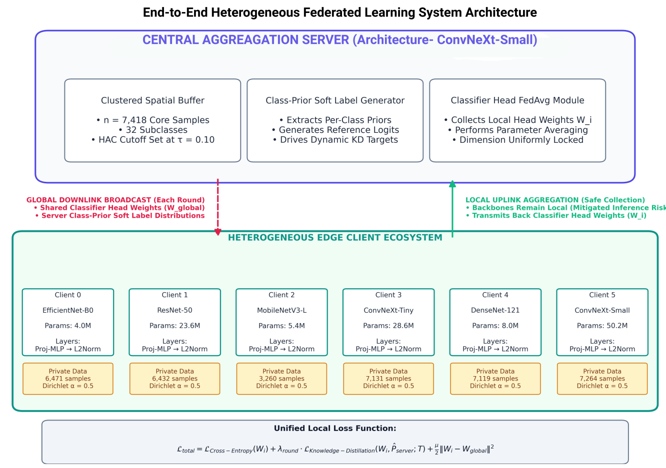

Fig. 1. Overall architecture of the proposed heterogeneous FL system. Six edge clients with distinct backbones communicate with the central server, which retains the clustered buffer, generates class-prior soft labels, and aggregates only the shared classifier head.

aggregation with personalised local adaptation to handle full heterogeneity [6]. In contrast, the proposed framework provides a *structural* privacy guarantee requiring no noise budget or cryptographic infrastructure.

D. Agricultural Pest Classification

Wu et al. introduced the IP102 benchmark with centralised deep feature baselines achieving 49.4-67.8% top-1 accuracy [4]. Mohsin et al. applied five CNNs with LIME-based explainability, reaching 67.5% with DenseNet-121 [17]. Both operate under centralised training with full dataset access; the proposed system achieves 63.76% under the strictly harder constraint of heterogeneous, non-IID, privacy-preserving federated training.

III. Proposed Methodology

A. System Overview and Design Rationale

The framework is designed around four objectives in descending order of precedence: (1) privacy by architectural design rather than noise injection; (2) cross-class generalisation beyond directly supervised buffer subclasses; (3) classifier accuracy exceeding the server-only baseline; and (4) architectural flexibility without disclosing client backbone choice. Standard homogeneous FedAvg fails the first two objectives outright, motivating the two-stage pipeline-offline buffer construction and online heterogeneous FL-shown in Fig. 1.

B. Dataset and Buffer Construction

All experiments use the IP102 benchmark [4]: 75,222 images across 102 fine-grained pest subclasses organised into 8 crop-host superclasses (Rice, Corn, Wheat, Beet, Alfalfa, Vitis, Citrus, Mango), split 45,095/7,508/22,619 for train/val/test. Prior to federation, a reference buffer is constructed as shown in Fig. 2. A pretrained ConvNeXt-Small backbone extracts 768-dimensional feature centroids for all 102 subclasses; a

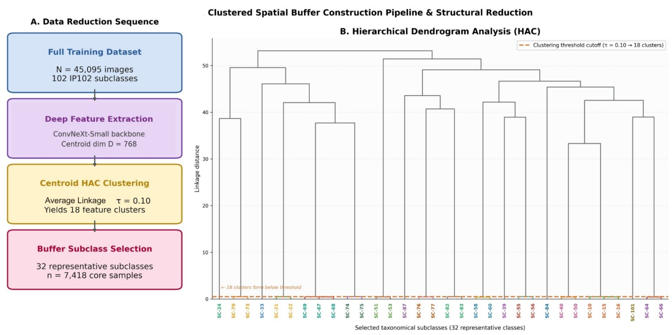

Fig. 2. Buffer construction pipeline. Left: the four-stage data reduction sequence from the full training set to the 7,418-sample buffer. Right: the HAC dendrogram over 102 subclass centroids; the cut at τ=0.10 yields 18 clusters.

TABLE I
BACKBONE RANKING BY BUFFER TEST ACCURACY.

Backbone	Feat. dim	Buf. acc.	Role
EfficientNet-B2	1,408	81.72%	Server
EfficientNet-B0	1,280	81.60%	Client 0
ResNet-50	2,048	80.44%	Client 1
MobileNetV3-Large	960	80.37%	Client 2
ConvNeXt-Tiny	768	79.02%	Client 3
DenseNet-121	1,024	78.80%	Client 4
ConvNeXt-Small	768	77.77%	Client 5

102×102 cosine distance matrix is computed and HAC with average linkage and threshold τ=0.10 yields 18 visually distinct clusters. From each cluster, 1-4 representative subclasses are selected proportional to cluster size, giving 32 buffer subclasses ($\mathcal{S}_{\text{buf}}$); a 20% stratified sample per subclass yields $|\mathcal{B}| = 7{,}418$ buffer images, excluded from all client Dirichlet partitions to guarantee zero server-client data overlap.

C. Unified Client and Server Architecture

A single model class, `FLClientModel`, serves all nodes. Backbone features of varying dimension d are projected through a shared two-layer bottleneck (hidden 512, output 256, BN + ReLU + Dropout p=0.5) and L2-normalised:

$$\mathbf{z} = \text{L2Norm}(\text{ProjMLP}(\text{Backbone}(\mathbf{x}))) \,, \tag{1}$$

$$\hat{\mathbf{y}} = \mathbf{W} \cdot \text{Dropout}(\mathbf{z}), \quad \mathbf{W} \in \mathbb{R}^{102\times256}. \tag{2}$$

The shared classifier head $\mathbf{W}$ (26,112 parameters) is the *only* component exchanged during aggregation; backbone and projection parameters remain local throughout training.

Seven ImageNet-pretrained backbones were ranked by buffer test accuracy (Table I). EfficientNet-B2 (81.72%) was assigned the server role, maximising the quality of the class-prior KD signal; the remaining six form the heterogeneous client ensemble.

Client data is partitioned with a Dirichlet protocol (α=0.5, superclass level): for each superclass, $D_i \sim \text{Dir}(\alpha \cdot \mathbf{1}_6)$ governs each client's allocation, preserving realistic within-farm specialisation. Per-client totals range from 3,260 to 7,264 private images, shown in Fig. 3.

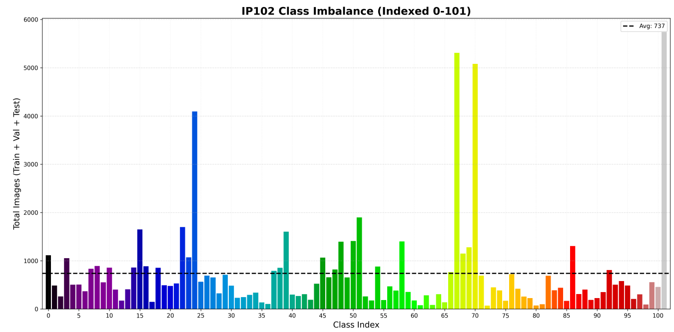

Fig. 3. Per-client private sample counts after Dirichlet partitioning ($\alpha = 0.5$, superclass level). Colour segments correspond to the 8 crop-host superclasses.

Algorithm 1 Heterogeneous FL with Server-Anchored Class-Prior KD

Require: Server model f_s, buffer $\mathcal{B}$, buffer subclasses $\mathcal{S}_{\text{buf}}$, client datasets $\{D_i\}_{i=1}^{6}$, R=100, $\lambda_{\min}$=0.30, $\lambda_{\max}$=0.90, $T_{\max}$=4.0, $T_{\min}$=2.5

Pretrain f_s on $\mathcal{B}$ for 50 epochs; save best-val checkpoint
Initialise global head $\mathbf{W}^{(0)} \leftarrow$ pretrained server head
for $t = 1, \ldots, R$ **do**
  $\lambda^{(t)} \leftarrow \lambda_{\min} + (\lambda_{\max} - \lambda_{\min})\frac{t-1}{R-1}$
  $T^{(t)} \leftarrow T_{\max} - (T_{\max} - T_{\min})\frac{t-1}{R-1}$
  Compute priors $\mathcal{P}^{(t)} = \{p_s^{(t)}\}_{s\in\mathcal{S}_{\text{buf}}}$ ▷ Eq. (3)
  for each client $i = 1, \ldots, 6$ **(in parallel) do**
    Receive $\mathbf{W}^{(t)}$ and $\mathcal{P}^{(t)}$ from server
    $\mathbf{W}_i^{(t+1)} \leftarrow \text{LocalTrain}(D_i, \mathbf{W}^{(t)}, \mathcal{P}^{(t)}, \lambda^{(t)}, T^{(t)})$ ▷ Eq. (7)
    Return $\mathbf{W}_i^{(t+1)}$ to server
  end for
  $\mathbf{W}^{(t+1)} \leftarrow \sum_{i=1}^{6} \frac{|D_i|}{\sum_j |D_j|} \mathbf{W}_i^{(t+1)}$
  Save checkpoint if best validation accuracy
end for
Load best-val checkpoint; evaluate ensemble on test set

D. Federated Learning Protocol

The complete protocol is formalised in Algorithm 1. It differs from standard FedAvg in three ways: (1) only the classifier head is aggregated; (2) the server broadcasts class-prior soft-label priors each round; and (3) a progressive schedule controls the distillation weight and temperature over the 100-round budget.

At the start of round t, the server averages per-subclass softmax outputs at temperature $T^{(t)}$:

$$p_s^{(t)} = \frac{1}{|\mathcal{B}_s|} \sum_{\mathbf{x}\in\mathcal{B}_s} \sigma\!\left(\frac{f_s(\mathbf{x})}{T^{(t)}}\right), \quad s \in \mathcal{S}_{\text{buf}}. \tag{3}$$

The resulting 32×102 class-prior matrix is broadcast alongside $\mathbf{W}^{(t)}$, reducing the distillation payload by 231× relative to per-image soft labels (Fig. 4).

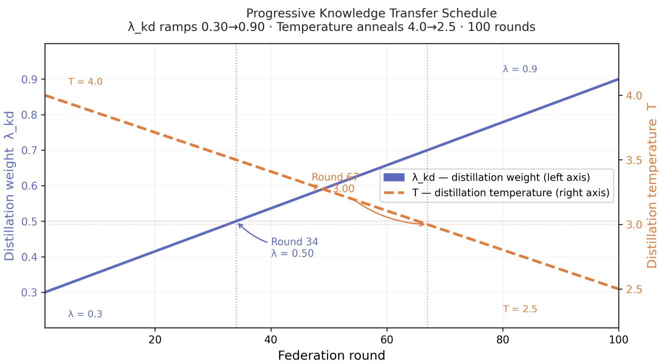

Fig. 4. Progressive KD schedule over 100 federation rounds. λ_{kd} ramps from 0.30 to 0.90 (left axis); temperature T anneals from 4.0 to 2.5 (right axis).

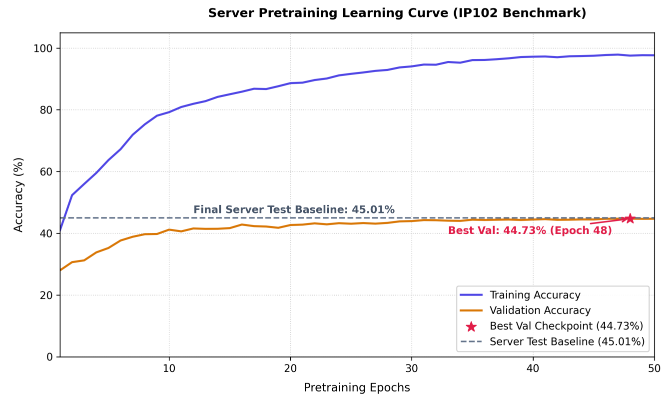

Fig. 5. Server pretraining learning curve for EfficientNet-B2 over 50 epochs. Best-val checkpoint (epoch 48, 44.73%) yields 45.01% test accuracy, the server-alone baseline.

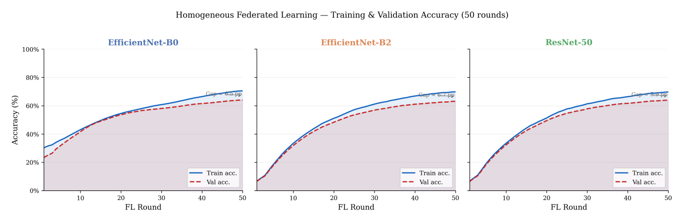

Fig. 6. Training and validation accuracy for the three homogeneous FL configurations (50 rounds). The small train-val gap confirms full-model sharing aligns client representations at the cost of gradient privacy.

E. Local Training Objective and Privacy Mechanism

Each client trains for one epoch per round. Let n_b, n_n be the buffer-/non-buffer-subclass sample counts in a mini-batch:

$$\mathcal{L}_{\text{CE}} = \frac{n_b \mathcal{L}_{\text{CE}}^{\mathcal{B}} + n_n \gamma\, \mathcal{L}_{\text{CE}}^{\mathcal{N}}}{n_b + n_n}, \quad \gamma = 1.5, \tag{4}$$

$$\mathcal{L}_{\text{KD}} = T^{(t)2} \cdot \frac{1}{n_b} \sum_{j \in \mathcal{B}_{\text{batch}}} \text{KL}\Big(\sigma\big(\tfrac{\mathbf{y}_j}{T^{(t)}}\big) \,\Big\|\, \sigma\Big(\tfrac{p_{s_j}^{(t)}}{T^{(t)}}\Big)\Big), \tag{5}$$

$$\mathcal{L}_{\text{prox}} = \tfrac{\mu}{2} \|\mathbf{W}_{\text{local}} - \mathbf{W}^{(t)}\|_F^2, \quad \mu = 0.01, \tag{6}$$

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{CE}} + \lambda^{(t)} \mathcal{L}_{\text{KD}} + \mathcal{L}_{\text{prox}}. \tag{7}$$

The amplification γ=1.5 counteracts the 1.9× loss-signal gap that non-buffer subclasses would otherwise face at peak λ; the FedProx term (6) prevents excessive single-epoch classifier drift.

The property inference attack is a cosine-similarity classifier: at round t, the server compares a client's head update $\Delta \mathbf{W}_i^{(t)}$ against eight superclass reference gradients $\{g_k\}_{k=1}^{8}$:

$$\hat{k}_i^{(t)} = \arg\max_k \frac{\Delta \mathbf{W}_i^{(t)} \cdot g_k}{\|\Delta \mathbf{W}_i^{(t)}\| \, \|g_k\|}. \tag{8}$$

In a homogeneous system, full model updates reside in a common space and (8) is well-defined. In the proposed system, backbone parameters are never transmitted; the only shared surface-the 26,112-parameter classifier head-carries no exploitable backbone-specific gradient structure, yielding identically 0% attack accuracy in every round.

IV. Experimental Results

A. Experimental Setup

Training augmentations include random resized cropping, random horizontal/vertical flip, RandAugment, ImageNet normalisation, and random erasing; evaluation uses deterministic resize-then-crop. All clients use AdamW (backbone LR $5{\times}10^{-5}$, head LR $5{\times}10^{-4}$, weight decay 10^{-3}, cosine annealing, batch size 128) on a single NVIDIA RTX 4080 SUPER (16 GB VRAM), seeded at 42. Ensemble inference averages six softmax vectors element-wise; the best-validation checkpoint is used for all test evaluations.

B. Server Baseline and Homogeneous FL

The EfficientNet-B2 server, pretrained standalone on the 7,418-sample buffer for 50 epochs, reaches **45.01%** test accuracy (82.20% buffer-class, 0.00% non-buffer-class), establishing the lower bound the federated ensemble must exceed (Fig. 5).

Three homogeneous FL runs (EfficientNet-B0/B2, ResNet-50) perform full-model FedAvg for 50 rounds, reaching 64.01%, 64.90%, and 65.75% test accuracy with a tight 6.0-7.2 pp train-val gap (Fig. 6). However, the property inference attack achieves **56.7-61.7%** accuracy-4.5-4.9× the 12.5% random baseline-confirming that full-model gradient sharing in a common parameter space is a severe, exploitable privacy vulnerability.

C. Heterogeneous FL: No-Buffer and Proposed System

Six heterogeneous clients without any buffer or KD signal achieve **59.60%** test accuracy (F1 = 0.533, 14.59 pp above server-only) with structurally **0%** attack accuracy, but only 33.14% non-buffer class accuracy and an 18.7 pp train-val gap, reflecting the absence of a shared knowledge anchor.

The proposed system-buffer retained at the server with the progressive KD schedule over 100 rounds-achieves **63.76%** test accuracy and macro-F1 **0.611**, the best F1 among heterogeneous configurations (Fig. 7). The key finding is the buffer/non-buffer asymmetry (Fig. 8): non-buffer accuracy improves by **12.57 pp** (33.14%→45.71%), 6.7× larger than

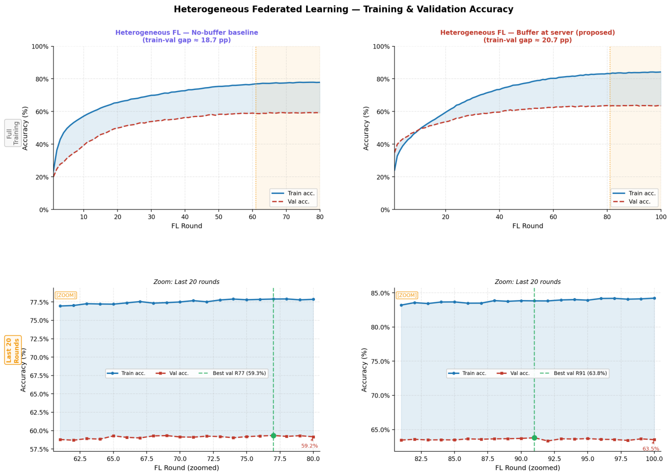

Fig. 7. Training/validation accuracy: no-buffer baseline (left) vs. proposed system (right). The widening gap on the right tracks the λ ramp, not overfitting. Best checkpoint: round 91 (63.80%).

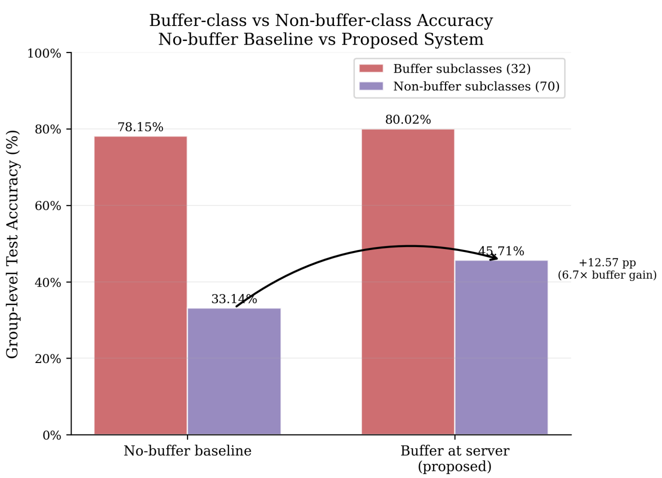

Fig. 8. Buffer-class vs. non-buffer-class test accuracy: no-buffer baseline vs. proposed system. The 6.7× asymmetry confirms cross-class coordination.

the buffer-class gain (1.87 pp), demonstrating genuine cross-class coordination rather than buffer-class specialisation.

D. Privacy Evaluation and Full Comparison

Fig. 9 plots attack accuracy per round for the homogeneous configurations; steady-state levels (56.7-61.7%) remain well above chance despite an 8-13 pp round-to-round fluctuation. All heterogeneous configurations record identically 0% with zero variance. Table II summarises all configurations: the proposed system is the only one to simultaneously achieve competitive accuracy (within 1.99 pp of the best homogeneous run), matching macro-F1 (0.611 vs. 0.612), non-trivial non-buffer accuracy, and zero attack accuracy.

E. Design Iteration and Per-Superclass Analysis

Table III tracks non-buffer accuracy across design adjustments. Relocating the buffer from clients to the server is the single most impactful change, converting a 5.03 pp regression

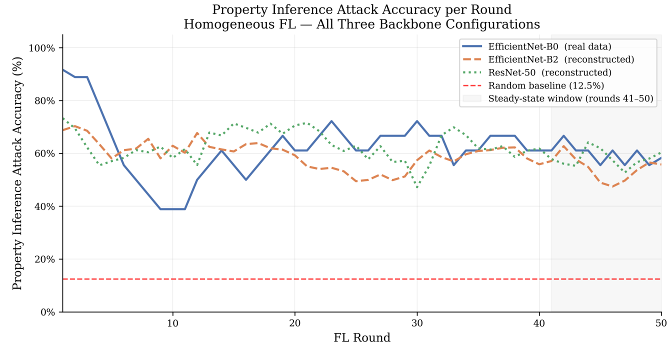

Fig. 9. Property inference attack accuracy per round for the homogeneous FL configurations (50 rounds), remaining 4.5-4.9× above the 12.5% random baseline.

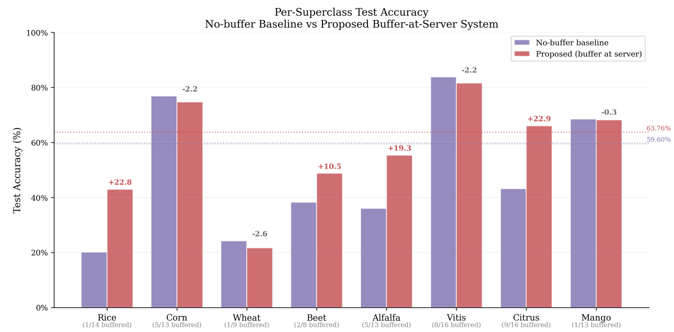

Fig. 10. Per-superclass test accuracy: no-buffer baseline (lighter) vs. proposed system (darker). Δ annotated above each pair.

into a 7.47 pp gain by eliminating within-batch gradient crowding; the progressive schedule adds a further 5.10 pp.

Fig. 10 and Table IV show three response patterns: *strong coordination* in Rice (+22.8 pp), Alfalfa (+19.2 pp), and Citrus (+22.9 pp), all starting below 45%; *near-saturation regression* in Corn (−2.3 pp) and Vitis (−2.2 pp), already above 80% pre-buffer; and *minimal response* in Wheat (−2.6 pp) and Mango (−0.3 pp), each with only one buffer subclass.

F. Comparison with Prior Work

Table V contextualises the proposed system against prior centralised IP102 work. The proposed system achieves 63.76% under a strictly harder constraint-fragmented, non-IID, never-centralised data with zero server inspection of client internals-only 3.75 pp below the centralised best (67.5%), consistent with the expectation that federated non-IID training incurs an accuracy penalty the buffer KD mechanism partially recovers.

V. Discussion

Cross-class coordination: The 6.7× ratio between the non-buffer gain (12.57 pp) and the buffer-class gain (1.87 pp) is the central empirical result. Eleven non-buffer subclasses transition from exactly 0% in the no-buffer baseline to non-zero accuracy (3.2-55.3%) in the proposed system, despite receiving no direct KD signal-evidence that the improvement originates from the structured initialisation of the shared

TABLE II
NUMERICAL COMPARISON OF ALL EXPERIMENTAL CONFIGURATIONS ON THE IP102 TEST SET. ATTACK BASELINE: 12.5% (RANDOM, 8-CLASS PROBLEM).

Configuration	Test acc.	F1	Buf. acc.	Non-buf. acc.	Attack acc.	Rounds
Server baseline (buffer only)	45.01%	0.337	82.20%	0.00%	-	50 ep.
Homogeneous FL (EfficientNet-B0)	64.01%	0.597	-	-	60.0%	50
Homogeneous FL (EfficientNet-B2)	64.90%	0.606	-	-	56.7%	50
Homogeneous FL (ResNet-50)	65.75%	0.612	-	-	61.7%	50
Heterogeneous FL, no buffer	59.60%	0.533	78.15%	33.14%	0.0%	80
Hetero FL, buffer at server (proposed)	**63.76%**	**0.611**	**80.02%**	**45.71%**	**0.0%**	**100**

TABLE III
DESIGN ITERATION HISTORY: NON-BUFFER ACCURACY RELATIVE TO THE NO-BUFFER BASELINE (33.14%).

Configuration	Non-buf. acc.	Δ
No-buffer baseline (reference)	33.14%	-
Buffer at clients, per-image KD	≈28.11%	−5.03 pp
Buffer at server, fixed $\lambda = 0.70$	40.61%	+7.47 pp
+ λ ramp + T anneal (100 rds)	**45.71%**	**+12.57 pp**

TABLE IV
PER-SUPERCLASS TEST ACCURACY FOR HETEROGENEOUS FL CONFIGURATIONS.

Superclass	Sub-cls	Buf. sub.	No-buf.	Proposed	Δ
Rice	14	1	20.2%	43.0%	+22.8 pp
Corn	13	5	76.9%	74.7%	−2.3 pp
Wheat	9	1	24.3%	21.7%	−2.6 pp
Beet	8	2	38.3%	48.8%	+10.5 pp
Alfalfa	13	5	36.1%	55.4%	+19.2 pp
Vitis	16	8	83.8%	81.6%	−2.2 pp
Citrus	16	9	43.2%	66.1%	+22.9 pp
Mango	13	1	68.5%	68.2%	−0.3 pp
Overall	**102**	**32**	**59.60%**	**63.76%**	**+4.16 pp**

classifier head rather than direct supervision. Alfalfa shows the clearest superclass-level pattern: non-buffer subclasses gain +26.8 pp versus +7.8 pp for buffer subclasses.

The privacy-accuracy tradeoff: The 1.99 pp accuracy gap between the proposed system and the best homogeneous configuration is the cost of *complete structural privacy protection*, not of heterogeneity itself-the homogeneous configuration's higher accuracy comes paired with 56.7-61.7% attack accuracy. In macro-F1 the gap is just 0.001. Communication cost is also far lower: the homogeneous configuration transmits up to 25.6 M parameters per client per round versus 26,112 (classifier head) plus a 3,264-float class-prior broadcast in the proposed system-roughly 980× less.

Structural limitations: The 18.7-20.7 pp train-val gap reflects the Dirichlet-skewed training distribution versus the balanced evaluation split, not memorisation: aggressive regularisation (MixUp, CutMix) closes the gap but costs 2.13 pp accuracy. Twenty-five subclasses remain at 0% accuracy in both configurations due to very low per-class image counts (mean 76 test samples) under Dirichlet partitioning, a limitation that future buffer-coverage extensions could address.

TABLE V
COMPARISON OF IP102 CLASSIFICATION APPROACHES.

Work	Method	Acc.	FL
Wu et al. (2019) [4]	Deep feature baselines	49.4-67.8%	No
Mohsin et al. (2022) [17]	DenseNet-121 + LIME-XAI	67.5%	No
Proposed	Hetero FL + buffer KD	**63.76%**	**Yes**

VI. CONCLUSION

This paper presented a heterogeneous federated learning framework for fine-grained agricultural pest classification on IP102, grounding privacy guarantees in system architecture rather than post-hoc noise injection. By assigning each of six clients a distinct backbone, the framework renders property inference attacks structurally impossible, reducing attack accuracy from 56.7-61.7% to 0% across every client and round. The server-retained HAC clustered buffer with class-prior knowledge distillation yields 63.76% test accuracy (macro-F1 = 0.611)-an 18.75 pp gain over the server-alone baseline and only 1.99 pp below the best homogeneous configuration-while non-buffer class accuracy improves by 12.57 pp (6.7× the buffer-class gain), confirming genuine cross-class coordination. Structural privacy is thus achievable at negligible accuracy cost, without differential privacy or cryptographic mechanisms. Future work includes adaptive buffer construction for currently unlearnable sparse subclasses, per-subclass distillation weights to mitigate ceiling-effect regression, multi-seed evaluation, and deployment across real network-connected devices.

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